

COLLECTIONS INTELLIGENCE

Unified Collections Intelligence Engine

iResolve Module Specification — Compliance, Propensity, RPC & Dispute Intelligence

PREPARED FOR Credence Resource Management (CRM) — iResolve Platform

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This document is an internal engineering proposal. Cost and ROI figures are illustrative planning estimates unless explicitly marked as sourced industry benchmarks — see Sections 10 and 11 for detail.

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1. Executive Summary

This document proposes a Unified Collections Intelligence Engine for iResolve — presented to end users as a single Modules Hub entry ("Collections Intelligence"), but built as four independently deployable services sharing one account-level data layer:

Compliance & Call Intelligence Layer — real-time FDCPA/Reg F/TCPA monitoring and gating of collector actions. Propensity-to-Pay + RPC (Right-Party-Contact) Scoring — predicts payment likelihood and contact reachability per account.

Dispute/Complaint NLP Triage — classifies inbound disputes and complaints, drafts compliant responses for human review.

Shared Account Feature Store — the connective data layer all three services read from and write to.

The system is designed to move CRM from static, aging-bucket-driven collections toward a scored, prioritized, and continuously compliance-checked workflow. Industry-wide AI/ML adoption in debt collection grew from roughly 49% of agencies in 2023 to about 93% in 2025, and published industry estimates put achievable operational cost reductions around 40% and recovery-rate improvements around 10% for agencies adopting propensity-based prioritization. These are industry-wide directional benchmarks, not CRM-specific guarantees — Section 11 explains how to convert them into a CRM-specific projection once real portfolio numbers are available.

Given CRM's documented FDCPA complaint history and a past Minnesota regulatory enforcement action, the recommended build order leads with the Compliance & Call Intelligence Layer — the piece with the clearest, fastest-to-justify business case and the least dependency on other systems — rather than starting with scoring models.

2. Business Problem & Opportunity

2.1 CRM's Collections Context

CRM operates as a first- and third-party ARM firm across telecommunications, utility, healthcare, insurance, retail, and financial services receivables, running its operations on the Artiva platform. These verticals behave differently:

Utility collections are seasonal — shutoff-notice cycles create predictable spikes in both payment likelihood and complaint sensitivity.

Healthcare collections are entangled with billing accuracy and insurance adjudication — a meaningful share of disputes here are legitimate billing errors, not payment avoidance.

Telecom collections tend to have higher account volume with lower average balances — the economics favor speed and automation over high-touch manual handling.

A single static aging-bucket strategy applied uniformly across these verticals leaves value on the table in two directions at once — under-prioritizing genuinely collectible accounts, and over-contacting accounts unlikely to pay or likely to generate a complaint. Neither shows up cleanly in a monthly liquidation report; both are only visible once accounts are scored individually.

2.2 The Gap

Industry data indicates that collection agencies typically recover only 20–30% of unpaid debts on average, with ineffective outreach, poor timing, and misaligned communication strategies cited as the primary causes. The probability of full recovery on a delinquent account falls sharply with time — roughly 69% at 3 months past due, 51% at 6 months, and 21% at 12 months per commercial collection industry studies — which means prioritization speed and accuracy compound directly into recovery outcomes.

This also has a second-order effect on collector experience: without scoring, collectors develop their own informal heuristics for which accounts feel worth working, which is inconsistent across a team, impossible to audit, and doesn't transfer when a collector leaves or a new hire joins. A shared, explainable score standardizes that judgment call across the floor.

2.3 The Regulatory Reality

CRM has a documented history of FDCPA/TCPA-related consumer complaints and at least one state-level (Minnesota) enforcement action. This is a material business risk independent of collections performance — complaint volume affects client trust, since CRM's clients are themselves exposed to reputational and regulatory risk through their vendor. A system that reduces complaint-generating contact before it happens is a distinct value stream from a system that improves liquidation — both matter, but should be built, measured, and justified separately.

2.4 Why Now

Two industry-level shifts make this the right moment to build this rather than defer it:

Regulatory tooling has matured. Real-time compliance monitoring has gone from a niche vendor category to a fairly standard expectation among ARM compliance teams over the last two years — building this in-house avoids being a laggard on what's increasingly treated as baseline practice.

The cost of the AI components has dropped enough that this is no longer a big-tech-only build. As Section 10 shows, the per-unit cost of the language-model components is small enough that the limiting factor is engineering time and data quality, not compute budget — that wasn't true even two or three years ago.

3. Regulatory & Compliance Context

Regulation	What it Constrains	Relevance to This System
FDCPA	Prohibits harassment, false representation, unauthorized third-party disclosure	Core rule set for the Compliance Layer's language/behavior checks
Reg F (CFPB)	Validation notice format; 7-in-7 call frequency rule	Directly enforceable as an automated gate on recommended contact frequency
TCPA	Consent requirements for autodialed/prerecorded calls and texts	Relevant to any automated outbound contact (e.g., SMS reminders)
State overlays	Many states add licensing, disclosure, and calling-hour rules stricter than federal floor	Rules engine needs a state-aware rule table, not a single national rule set
HIPAA	PHI protection for healthcare-vertical accounts	All services must route healthcare-vertical data through CRM's existing PHI scrubber

The industry direction as of 2026 is toward continuous, real-time compliance monitoring rather than after-the-fact sampling. This proposal follows that established pattern rather than inventing a novel compliance approach.

4. Proposed Solution — Four Services, One Product Surface

4.1 Compliance & Call Intelligence Layer (build first)

Transcribes collector calls (ASR) and analyzes SMS/email content.

Checks every planned or completed contact against a rules table: mini-Miranda disclosure, prohibited language, Reg F 7-in-7 cap, permitted calling hours, state overlays, unauthorized disclosure.

Operates as a gate, not an advisor — a violating recommended action is automatically downgraded or blocked, with the reason logged.

Every decision is logged with a rule ID, timestamp, and account ID for CFPB or internal legal review.

AI models used, and why

Component	Model	Why This Model
Speech-to-text (ASR)	Whisper (open-source, self-hosted) — medium or small.en variant	Zero recurring API cost, runs on existing HIPAA server, no call audio leaves CRM infrastructure
Objective rule checks (frequency, hours, disclosure)	Deterministic rules engine, not an ML model	Checkable facts, not judgment calls — cheaper and more defensible to a regulator
Nuanced language flagging (tone, false representation)	Claude Haiku 4.5, via Anthropic API, Batch mode	Anthropic's fastest, lowest-cost model, built for high-volume classification; batch mode halves cost since near-real-time is sufficient

How it works, step by step

1. A collector places or receives a call in Airtiva (existing call recording infrastructure captures the audio, unchanged).
2. Post-call, the recording is queued as a Celery job and transcribed by Whisper, speaker-diarized where possible (collector vs. debtor turns).
3. The transcript runs through two parallel checks: (a) deterministic rule checks for mini-Miranda presence, forbidden phrases, Reg F 7-in-7 count, and calling-hour rules; (b) Claude Haiku classifies ambiguous language (tone, implied threats, false representation) a keyword rule would miss.
4. Results write to `compliance_gate_result` in the feature store. Anything flagged routes to a supervisor review queue with the relevant transcript excerpt and reason attached.
5. Separately and faster (synchronous, sub-second): before any recommended action from the propensity engine reaches a collector's screen, the same rule set checks the planned action itself and downgrades or blocks it if it would violate a rule — this is the real-time gate function.

Business benefit

Reviews 100% of calls instead of a small manual sample — catches violations currently invisible to spot-check QA, directly reducing FDCPA/TCPA complaint exposure.

A fully logged, rule-traceable decision trail cuts the time needed to respond to a regulator inquiry or lawsuit discovery request.

Generates structured coaching data per collector as a byproduct, improving call quality over time without a separate QA program.

The real-time gate prevents violations before they happen, rather than only detecting them after the fact.

4.2 Propensity-to-Pay + RPC Scoring

Propensity-to-Pay: predicts probability of payment in 7/30/60-day windows using balance, aging, vertical, and contact/promise-to-pay history.

RPC confidence: scores each phone/email/address by likelihood of reaching the actual debtor; feeds into propensity as a feature.

Build-vs-buy note: confirm whether CRM already licenses skip-trace/contact data (e.g., LexisNexis, TLO) before building an in-house RPC model — a vendor feed may outperform an in-house model trained only on CRM's own history.

Caution: a Yale-affiliated study found borrowers repaid less when an AI made first contact rather than a human — this system should surface recommendations to collectors, not autonomously initiate contact, in its initial phases.

AI models used, and why

Component	Model	Why This Model
Propensity-to-pay scoring	XGBoost / LightGBM, self-hosted, trained on Artiva history	Tabular data is exactly what gradient-boosted trees are built for — explainable via SHAP, trains in minutes on CPU, no LLM or API cost
RPC confidence scoring	Logistic regression / gradient-boosted trees on dial-outcome labels	Probability estimation on structured data, not a language task — no LLM needed

How it works, step by step

1. A scheduled Celery job (nightly, or more frequent for high-priority segments) pulls the latest account state from Artiva into the shared feature store — balance, aging, vertical, payment/promise-to-pay history.
2. A feature engineering step derives secondary features: days since last payment, promise-to-pay keep ratio, count/outcome of prior contact attempts, seasonality flags.
3. The XGBoost/LightGBM propensity model scores every open account for 7/30/60-day payment probability. The RPC model separately scores each phone/email/address for reachability.
4. A decision layer combines both scores into a recommended action — e.g., high propensity + high RPC confidence -> call now with a specific offer; high propensity + low RPC confidence -> attempt skip-trace or switch channel; low propensity -> deprioritize.
5. The recommended action passes through the Compliance Layer's real-time gate before it's surfaced on the collector's Artiva screen via the existing COS bridge.
6. Whatever the collector actually does gets logged back to contact_history; the propensity model retrains monthly (or on drift) using this growing outcome history — a genuine feedback loop.

Business benefit

Directly targets liquidation rate — the primary KPI CRM's clients score the agency on — by pointing collector effort at accounts statistically likely to pay.

RPC scoring reduces wasted dial attempts on unreachable numbers, increasing completed contacts per collector-hour without adding headcount.

SHAP-based explainability means a prioritization decision can be justified to a client or auditor, not just a black-box output.

Segment-level score trends give leadership an early, data-backed signal on where collection strategy needs to change.

4.3 Dispute/Complaint NLP Triage

Classifies inbound letters, emails, and complaints by type and urgency.

Routes to a human-reviewed dashboard with a drafted, compliance-checked response — not autonomous response-sending.

Feeds aggregated, periodic trend data into strategy reviews only. Individual dispute events are not used as a live propensity-model training label — they are too sparse and causally confounded to be a safe signal without separate causal analysis.

AI models used, and why

Component	Model	Why This Model
Dispute type/urgency classification	Self-hosted ModernBERT + LoRA multi-label classifier	Closed-set categorization — a small fine-tuned encoder is accurate and cheap, with zero per-request cost once trained
Draft response generation	Claude Sonnet (claude-sonnet-5), synchronous, human-reviewed	Genuinely needs generative language quality for legally sensitive text — every draft reviewed before sending

How it works, step by step

1. Inbound correspondence — mailed letters (scanned/OCR'd), emails, or debtor portal submissions — lands in a triage queue.
2. The self-hosted ModernBERT+LoRA classifier tags each item with a dispute type and urgency score.
3. High-urgency or legally sensitive items (e.g., attorney letters, cease-and-desist notices) route immediately to a human reviewer, bypassing auto-drafting entirely.
4. For standard categories, Claude Sonnet drafts a response using account context pulled from the shared feature store (balance, history, applicable regulation).
5. The draft lands in a compliance/legal review queue; a human edits and approves before anything is sent — mirroring the BRD Interpreter's confirmation-gate pattern.
6. The dispute event and category are logged to `dispute_flags[]` — feeding only aggregated, periodic strategy review, not live model training.

Business benefit

Cuts response drafting time substantially, freeing compliance staff for judgment-heavy review rather than repetitive writing.

Faster turnaround reduces the risk of missing statutory response deadlines (e.g., the 30-day FD CPA validation window).

Consistent, pre-vetted response language reduces variance in quality/tone across different staff members.

The aggregated trend view gives compliance a proactive early-warning signal on problem scripts/segments before they escalate to a regulator.

4.4 Shared Account Feature Store

A PostgreSQL-based store of account-level features (contact history, RPC confidence, dispute flags, outcome labels, vertical, aging, balance). All three services read from and write to it via simple APIs, not a shared

codebase — this is what lets the services ship, fail, and be maintained independently while still working together. No AI model runs here — it's a data layer, not an inference service.

How it works, step by step

1. Scheduled Celery ETL jobs sync relevant Artiva data into PostgreSQL tables on a defined cadence (nightly for bulk fields, more frequent for high-priority accounts).
2. Each of the three services reads and writes through internal FastAPI endpoints, not direct database access, so permission boundaries are enforced per service even though the data lives in one place.
3. A retention/purge policy (defined with legal) governs how long call transcripts, dispute correspondence, and scores are kept.

Business benefit

Avoids three separate, redundant ETL pipelines being built and maintained independently — a real engineering-time saving over the system's life.

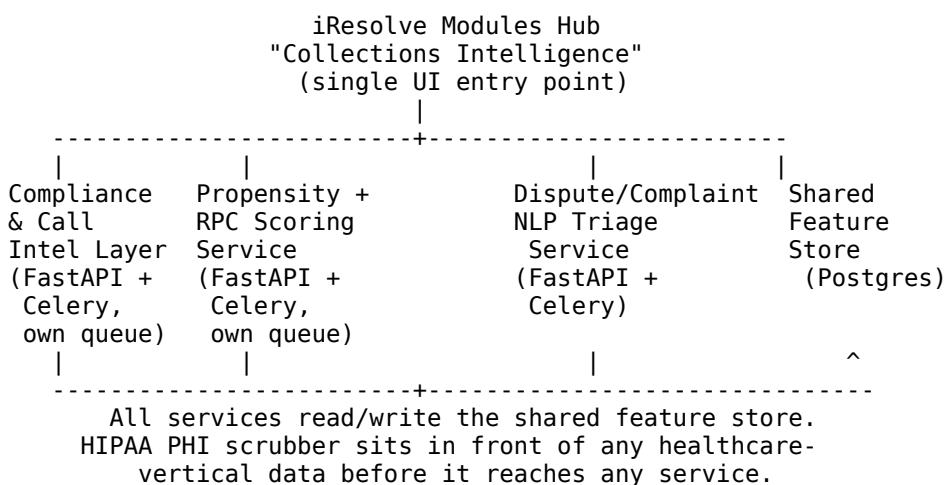
Enables genuinely cross-system features (e.g., propensity using RPC confidence and dispute history as inputs) without tightly coupling codebases.

Centralizes HIPAA/PHI governance to one data layer instead of enforcing it three separate times.

4.5 Why So Little of This Actually Needs an LLM

Three of the four services (compliance rule-checking, propensity scoring, RPC scoring) are primarily classical ML or deterministic logic, not LLM calls. LLMs (Haiku, Sonnet) are used only where the task is genuinely a language-understanding or language-generation problem — nuanced violation phrasing and dispute-response drafting. This matters directly for Section 10's cost analysis: most of this system's compute cost is effectively free (self-hosted, CPU-based), and the parts that do call the Claude API are narrow, low-token-volume tasks, not continuous LLM usage.

5. System Architecture



Stack Alignment with Existing iResolve Infrastructure

FastAPI + PostgreSQL + Celery/Redis — matches the RCA/CAPA and QA module pattern already in production. IIS reverse proxy, ADFS SSO, Windows Auth/LDAP group permissions — reused as-is, with a distinct permission boundary per service.

Self-hosted, zero/low recurring-cost preference (matching the SAST/DAST module constraint) — open-source Whisper for transcription and open-source classifiers for dispute triage by default, unless a HIPAA-compliant BAA-covered API is specifically approved.

6. Data Schema (Shared Feature Store — Core Fields)

Field	Description	Written By
account_id	Primary key, links to Artiva account	—
vertical	telecom / utility / healthcare / retail / financial	ETL from Artiva
balance, aging_days	Standard collections fields	ETL from Artiva
contact_history[]	Timestamped log of contact attempts, channel, outcome	Compliance Layer, collectors
rpc_confidence	0–1 score per contact point	Propensity/RPC Service
propensity_score	0–1, per time horizon	Propensity Service
recommended_action	Channel, timing, offer type	Propensity Service
compliance_gate_result	Approved / downgraded / blocked + rule ID	Compliance Layer
dispute_flags[]	Type, date, status	Dispute Triage Service
promise_to_pay_history[]	Made vs. kept	ETL from Artiva

7. Model Design

7.1 Propensity-to-Pay

Algorithm: Gradient-boosted trees (XGBoost or LightGBM) over a deep-learning approach — tabular data, need for feature-level explainability. Retraining cadence: monthly, or triggered by performance drift. SHAP values logged per prediction so any scored account's reasoning can be reconstructed on demand.

7.2 RPC Confidence

Logistic regression or gradient-boosted trees on historical dial-outcome labels. Evaluate against any existing vendor skip-trace feed before committing engineering time.

7.3 Dispute/Complaint NLP Triage

Fine-tuned or prompted classification (reusing the existing ModernBERT/LoRA multi-label classification pattern) over dispute categories, with a human-in-the-loop confirmation gate before any drafted response is sent — mirroring the BRD Interpreter's existing pattern.

7.4 Compliance Rules Engine

Deterministic rules table, not a model, for auditability and legal defensibility. ASR transcript to keyword/intent classifier flags potential violations for supervisor review; flags do not need to be 100% precise since a human reviews every flag before any corrective action.

8. Phased Build Roadmap

Phase	Deliverable	Rationale for Sequencing	Est. Duration
1	Compliance & Call Intelligence Layer	Fastest, clearest ROI; addresses documented complaint/enforcement history; no dependency on other phases	6–8 weeks
2	Shared Account Feature Store	Infrastructure needed by every later phase; low risk, high reuse value	3–4 weeks (overlaps Phase 1)
3	Propensity-to-Pay + RPC Scoring	Needs Phase 2's data layer; highest engineering complexity	8–10 weeks
4	Dispute/Complaint NLP Triage	Lower urgency than Phase 1; benefits from Phase 1's compliance taxonomy	5–6 weeks

Total estimated timeline: roughly 5–6 months for all four phases with a small team, assuming no major infrastructure blockers.

8.1 Milestone Detail Per Phase

Phase 1 — Compliance & Call Intelligence Layer

Weeks 1-2: Rule catalog definition with legal/compliance (FDCPA, Reg F, TCPA, state overlays); Whisper deployment and transcription accuracy testing on real call samples.

Weeks 3-5: Deterministic rule engine build, Claude Haiku integration for nuanced language flagging, supervisor review dashboard.

Weeks 6-8: Real-time gate integration with the collector's Artiva screen; pilot on one collector team, tuning false-positive rate before full rollout.

Phase 2 — Shared Account Feature Store

Weeks 1-2: Schema design, ETL job design against Artiva.

Weeks 3-4: Access-control boundaries per consuming service, retention policy sign-off with legal.

Phase 3 — Propensity-to-Pay + RPC Scoring

Weeks 1-2: Build-vs-buy decision on RPC (vendor feed vs. in-house), historical data extraction and labeling.

Weeks 3-6: Model training, SHAP explainability validation, backtesting against historical liquidation outcomes.

Weeks 7-10: Recommended-action decision layer, compliance-gate integration, pilot on one portfolio vertical before cross-vertical rollout.

Phase 4 — Dispute/Complaint NLP Triage

Weeks 1-2: Dispute taxonomy definition (reusing Phase 1's compliance categories where they overlap), classifier training on historical correspondence.

Weeks 3-4: Claude Sonnet drafting integration, human-review dashboard.

Weeks 5-6: Pilot with compliance/legal staff, response-quality review before full handoff.

9. Team & Resourcing

Role	Phase 1	Phase 2	Phase 3	Phase 4
Data/AI Engineer (lead)	Full-time	Full-time	Full-time	Full-time
Backend/Full-stack support	Part-time	—	Part-time	Part-time
Compliance/Legal reviewer (existing staff)	At rule-definition & go-live	—	On offer-bound rules	On response templates

No new headcount is assumed to be required if staffed as an extension of current iResolve development responsibilities, with periodic legal/compliance review sessions rather than a dedicated compliance engineer.

10. Cost Analysis — AI Usage Cost vs. Time-Cost Benefit

This section deliberately excludes internal engineering labor cost (sunk cost if built as part of existing iResolve development work) and third-party vendor subscriptions (those need real vendor quotes). What follows is what's actually quantifiable now: what the AI models cost to run, versus the staff time they free up. Volume assumptions are illustrative placeholders — swap in CRM's real monthly volumes for an exact figure — but the unit costs (Anthropic API rates) are real, current published rates.

10.1 What Actually Costs Money to Run

Most of this system is self-hosted classical ML with zero recurring cost — Whisper for transcription, XGBoost/LightGBM for propensity and RPC scoring, and a self-hosted ModernBERT/LoRA classifier for dispute categorization all run on existing server compute with no per-request bill. The only components that call a paid API are the two language-heavy tasks: nuanced compliance-language flagging (Claude Haiku) and dispute response drafting (Claude Sonnet).

Current published Anthropic API rates (per million tokens): Claude Haiku 4.5 — \$1.00 input / \$5.00 output. Claude Sonnet — \$3.00 input / \$15.00 output. Batch API halves both.

10.2 Illustrative Monthly AI Usage Cost

Task	Model	Assumed Monthly Volume*	Monthly Cost
Nuanced call-language flagging	Claude Haiku 4.5, Batch (50% off)	60,000 calls, ~1,500 in/100 out tokens	~\$60/month
Dispute response drafting	Claude Sonnet, synchronous	500 disputes, ~1,000 in/500 out tokens	~\$5-8/month
Total estimated AI API cost			~\$65-70/month (~\$800/year)

**Volumes are illustrative placeholders — replace with CRM's actual monthly call and dispute counts. The point that holds regardless of real volume: at these token rates, even 5-10x these volumes stays in the low hundreds of dollars per month, not thousands.*

Add a one-time, non-recurring hardware allowance of roughly \$0-3,000 only if the existing HIPAA server needs a modest GPU add-on for faster Whisper transcription at higher call volume — otherwise this is a near-zero infrastructure cost system.

10.3 Time-Cost Benefit (Staff Hours Freed)

Activity Today (Manual)	Activity With AI Assist	Illustrative Time Saved
Compliance QA samples ~5% of calls for review, missing most calls entirely	AI reviews 100% of calls; humans review only the ~3-5% flagged	Redirects reviewer time from blind sampling to targeted review
Staff draft each dispute response from scratch (~20-30 min each)	Staff review/edit an AI-drafted response (~5-10 min each)	~15-20 min saved per dispute x 500/month ~= 125-165 hours/month
Collectors dial numbers with no reachability signal	RPC confidence lets collectors skip low-probability contacts	Fewer wasted dial attempts, more completed contacts per collector-hour

Illustrative value of time saved: using a placeholder loaded staff cost of \$10/hour (adjust to CRM's actual blended rate), roughly 150 hours/month saved on dispute drafting alone is worth roughly \$1,500/month (~\$18,000/year) — against an AI API bill of roughly \$65-70/month (~\$800/year). That's before counting compliance QA time freed or collector-dial-efficiency gains, both of which add to the same side of the ledger.

10.4 The Honest Takeaway

The AI usage cost here is genuinely small — a few hundred to low-thousands of dollars a year, not tens of thousands — because the design deliberately keeps LLM usage narrow (Section 4.5) and pushes everything else to free, self-hosted classical ML. The real cost of this project is engineering time to build it, not AI usage to run it — that's a separate, one-time internal-labor conversation to have directly using CRM's actual cost figures, not something this document should guess at. What this section shows is that once built, the system is cheap to operate relative to the staff time it frees up.

11. Business Benefit / ROI Analysis

11.1 Why a Precise ROI Number Isn't Given Here

A real dollar ROI depends on CRM's actual placed-account volume, average balance, current liquidation rate, and current complaint-related cost exposure — figures internal to CRM's finance and legal functions not available for this document. What follows is a framework and worked illustrative example, not a claim about CRM's actual expected returns.

11.2 ROI Framework

Liquidation lift (Propensity + RPC): Additional annual recovery $\approx$ Total annual placed receivables $\times$ liquidation lift %.

Industry-cited directional benchmarks suggest AI-driven prioritization can lift recoveries by roughly 10% and reduce operational costs by up to 40% for agencies making full use of propensity scoring and automation — figures observed across the broader industry, not specific to CRM.

Illustrative example only (placeholder portfolio size): if CRM places \$50M in annual receivables and currently liquidates at a 25% rate (\$12.5M recovered), even a conservative 2–3% relative lift in liquidation (well below the ~10% industry benchmark) would represent roughly \$250,000–\$375,000 in additional annual recovery. Replace the \$50M placeholder with CRM's actual figure for a real projection.

Illustrative Three-Scenario View

Same \$50M/25% placeholder portfolio, purely to show how the range moves — replace with real figures:

Scenario	Liquidation Lift Assumption	Additional Annual Recovery (Illustrative)
Conservative	+2% relative lift	~\$250,000
Moderate	+5% relative lift	~\$625,000
Optimistic (in line with industry benchmark)	+10% relative lift	~\$1,250,000

Cost-per-contact reduction (RPC): reducing wasted dial attempts on unreachable numbers by 15–20% translates into freed collector capacity or reduced cost per completed contact — quantifiable once CRM's current dials-per-RPC figure is available.

Complaint/legal risk reduction (Compliance Layer): each substantiated FDCPA/TCPA complaint carries direct cost (legal response, potential settlement, potential regulatory action) and indirect cost (client relationship risk). Given CRM's documented complaint and enforcement history, even a modest reduction in complaint-generating contact has asymmetric value relative to its engineering cost — the strongest downside-protection argument in this proposal, independent of any liquidation upside.

11.3 Non-Financial Benefits

Auditability: a fully logged, rule-traceable compliance decision trail is valuable during any future regulatory inquiry, independent of whether it prevents a specific incident.

Client trust: telecom/utility/healthcare clients are increasingly likely to ask about AI governance and compliance monitoring — having this system in place is a competitive/retention factor for client relationships.

Collector experience: propensity-ranked queues and clear compliance guardrails reduce collector guesswork and are associated with reduced hiring/retention pressure industry-wide.

12. Risks & Mitigation

Risk	Mitigation
ASR/NLP misclassifies a compliant call as a violation	All flags route to human supervisor review before any action is taken
Propensity model treated as a black box by legal/compliance	Explainable gradient-boosted trees + SHAP logging, not deep learning
RPC model underperforms a vendor data feed	Explicit build-vs-buy checkpoint before Phase 3 engineering begins
Dispute data leaking into propensity training causes bias	Architectural decision to keep dispute data as aggregate strategy input only
HIPAA/PHI exposure for healthcare-vertical	All services route through the existing PHI scrubber before

Risk	Mitigation
accounts	reaching any model or log
Scope creep — one service's delay blocking another	Services independently deployable; Phase 1 ships with standalone value regardless of later phases

13. Success Metrics (KPIs)

Metric	Owner	Target Direction
FDCPA/Reg F/TCPA complaint rate (per 1,000 contacts)	Compliance Layer	Decrease
Reg F 7-in-7 violations caught pre-contact	Compliance Layer	Track, trend toward zero
Liquidation rate (overall and by vertical)	Propensity Service	Increase
Dials-per-right-party-contact	RPC Service	Decrease
Average dispute resolution time	Dispute Triage	Decrease
Collector adoption rate of recommended actions	All	Increase

14. Appendix

14.1 Glossary

RPC — Right-Party-Contact: successfully reaching the actual debtor.

Reg F 7-in-7 rule — CFPB rule limiting a collector to 7 calls in 7 days per debt, with no call within 7 days of a conversation about that debt.

Propensity-to-pay — Predicted probability that a given account will make a payment within a defined time window.

NBA (Next-Best-Action) — The recommended channel, timing, and offer for a given account, generated from the propensity model.

14.2 Sources Referenced (Industry Benchmarks, General — Not CRM-Specific)

Industry AI/ML adoption trend in debt collection (49% in 2023 to 93% in 2025) and related 2026 forecasting commentary.

McKinsey-cited operational cost reduction (~40%) and recovery-rate improvement (~10%) benchmarks for AI-adopting collection agencies.

Yale-affiliated research finding on reduced repayment when AI initiates first contact.

CFPB Regulation F requirements (7-in-7 call frequency rule, validation notice format).

Publicly available information on CRM's business lines and documented complaint/regulatory history, gathered via public web sources for context in Sections 2–3.

All figures outside of Section 14.2's cited industry benchmarks are engineering-effort estimates and illustrative placeholders, and should be treated as a planning starting point, not a committed budget or guaranteed return.